Final Exam

Name: ____________________

Please circle your section number!

Section 1	Section 2	Section 3	Section 4
Chenxi Wu (TR 1:00 PM)	Niclas Technau (MWF 1:20 PM)	Ander Aguirre (TR 9:30 AM)	Sarah Tammen (MWF 12:05 PM)

- **There are 7 problems on the exam, some of them have multiple parts.**
- Read the problems carefully and **budget your time wisely**.
- If you are running out of time, please at least say *how* you would solve a given problem, preferably using notation, even if you do not have the time to do so. This may earn you partial credit.
- No calculators or other electronic devices, please. **Turn off your phone.**
- You do not have to carry out complicated numerical computations, but you should simplify your answer if it is possible with reasonable effort. In particular, geometric series must be evaluated.
- Please present your solutions in a clear manner. **Justify your steps.** Numerical answers without explanation cannot get credit. Cross out the writing that you do not wish to be graded on.
- You can use the empty pages for additional space for writing if necessary.
- **Throughout** this exam $X, X_1, X_2, \dots$ and Y denote random variables.

Question	Points	Score
1	25	
2	20	
3	20	
4	11	
5	20	
6	19	
7	20	
Total:	135	

1. Suppose that X, Y are jointly continuous random variables with joint probability density function $f(x, y)$. Assume that $f(x, y) > 0$ for each $x, y \in \mathbb{R}$.

   Give **an expression** in terms of $f(x, y)$ for each of the following quantities.

   Note: Throughout this problem you do not need to and are advised not to justify your answer. (Pay attention to the endpoints of any integration you may write down, and notice that X, Y might not be independent.)

   (a) (5 points) The probability density function of Y.

   (b) (5 points) The probability $P(X + Y < 5)$.

   (c) (5 points) The expectation of X^3Y^2.

(d) (5 points) The conditional probability density of X given $Y = y$.

(e) (5 points) $E[X^3|Y = y]$.

2. Throughout this problem X, Y denote distinct (not necessarily independent) random variables. Further, $X_1, \ldots, X_n$ denotes a family of independent and identically distributed continuous random variables.

   Note: For part (a) you do not need to and are advised not justify your answer.

   (a) (5 points) Find the probability that $X_1 > X_2 > \cdots > X_n$.

   (b) (8 points) Show $\mathrm{Cov}(X, -X) \leq 0$. Further, show $\mathrm{Cov}(X, -X) = 0$ if and only if $P(X = E[X]) = 1$.

   (c) (7 points) Let X, Y be independent and identically distributed. Verify that $X + Y$ and $X - Y$ are uncorrelated.

3. This problem has two parts.

   (a) (10 points) Let $X, Y \sim \text{Unif}[-10, 10]$ be independent. Find the probability density function of $X+Y$.

(b) (10 points) Let $X \sim \text{Poisson}(2)$ and $Y \sim \text{Poisson}(7)$ be independent. Verify $X + Y \sim \text{Poisson}(9)$.

4. (11 points) Let X be a random variable with probability density function $f_X(x) = \frac{e^x}{e-1}$ on $[0, 1]$ and $f_X(x) = 0$ otherwise. Compute the moment generating function M_X of X.

5. A group of 40 students does a Secret Santa gift exchange, in which each student writes their name on a slip of paper, the slips are put into a bucket, and each student draws a name. Each student then buys a gift for the person whose name they drew. Let X be the number of students who drew their own name.

   (a) (7 points) Use the Principle of Inclusion and Exclusion to write a sum giving the probability that at least one person gets their own name. (There is no need to evaluate the sum.)

   (b) (5 points) Compute the expected value of X by writing X as a sum of indicator random variables.

(c) (8 points) Compute the variance of X.

6. Let $X_1, X_2, \ldots$ be independent and identically distributed random variables. Suppose $X_1 \sim \text{Unif}([-1,1])$. Define

$$Y_n = \frac{1}{\sqrt{n}} \sum_{i \leq n} X_i.$$

In some parts of this problem, you may wish to use the fact that if $U \sim \text{Unif}([a,b])$, then $\text{Var}(U) = \frac{1}{12}(b-a)^2$.

(a) (7 points) Use Chebyshev's inequality to find a upper bound of the probability that $Y_n > 1$.

(b) (9 points) What is the Central Limit Theorem estimate for this probability as $n \to \infty$

(c) (3 points) A non-negative random variable has $E[X] = 2$, $E[X^3] = 9$ and $E[X^4] = 20$. With this information what is the best bound you can give on $P(X \geq 5)$?

7. Let $X, Y \sim \text{Exponential}(\mu)$ be independent random variables. Let $Z = X + Y$.

   (a) (10 points) Find the conditional density function of X given $Z = z$.

(b) (10 points) Find the conditional expectation $E[X|Z = z]$.

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List of named distributions

$X \sim \text{Bernoulli}(p)$, PMF: $p_X(0) = 1 - p, \quad p_X(1) = p$

$X \sim \text{Binomial}(n, p)$, PMF: $p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \le k \le n$

$X \sim \text{Geometric}(p)$, PMF: $p_X(k) = p(1-p)^{k-1}, \quad k = 1, 2, \ldots$

$X \sim \text{Poisson}(\lambda)$, PMF: $p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \ldots$

$X \sim \text{Hypergeometric}(N, N_A, n)$, PMF: $p_X(k) = \frac{\binom{N_A}{k}\binom{N-N_A}{n-k}}{\binom{N}{n}}, \quad 0 \le k \le n$

$X \sim \text{Uniform}(a, b)$, PDF: $f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b], \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Normal}(\mu, \sigma^2)$, PDF: $f_X(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}$

$X \sim \text{Exponential}(\lambda)$, PDF: $f_X(t) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Gamma}(r, \lambda)$, PDF: $f_X(t) = \begin{cases} \frac{1}{\Gamma(r)} \lambda^r x^{r-1} e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$

$(X_1, \ldots, X_k) \sim \text{Multinomial}(n, k, p_1, \ldots, p_k)$,
joint PMF: $p(a_1, \ldots, a_k) = \binom{n}{a_1, \ldots, a_k} p_1^{a_1} \cdots p_k^{a_k}$ for $a_1, \ldots, a_k \ge 0$ with $a_1 + \cdots + a_k = n$.

The following table shows approximate values of the cumulative distribution function of the standard normal distribution.

x	$\Phi(x)$
-2.0	0.022
-1.9	0.028
-1.8	0.035
-1.7	0.044
-1.6	0.054
-1.5	0.066
-1.4	0.080
-1.3	0.096
-1.2	0.115
-1.1	0.135
-1.0	0.158
-0.9	0.184
-0.8	0.211
-0.7	0.242
-0.6	0.274
-0.5	0.308
-0.4	0.344
-0.3	0.382
-0.2	0.420
-0.1	0.460
0.0	0.500
0.1	0.539
0.2	0.579
0.3	0.617
0.4	0.655
0.5	0.691
0.6	0.725
0.7	0.758
0.8	0.788
0.9	0.815
1.0	0.841
1.1	0.864
1.2	0.884
1.3	0.903
1.4	0.919
1.5	0.933
1.6	0.945
1.7	0.955
1.8	0.964
1.9	0.971
2.0	0.977